SESSION 4

Wildcards & Pattern Matching

Full Questions • SQL Answers • Easy English Explanations

Session Topics

- LIKE — match a text pattern, such as names ending with 'Cafe'.
- BETWEEN — filter values within a range; both endpoints are included.
- IN — check whether a value matches any value in a list.
- Wildcard % — represents zero or more characters.

Learning goal: Use LIKE, BETWEEN, and IN to filter rows based on text patterns, numeric ranges, and multiple possible values.

Task 1 — Restaurants ending with “Cafe”

Question: Write an SQL query to find all restaurants in a table called Restaurants whose names end with 'Cafe' using the LIKE operator.

SQL Answer

```
SELECT *
FROM Restaurants
WHERE restaurant_name LIKE '%Cafe';
```

Explanation

- SELECT * displays all columns for each matching restaurant.
- FROM Restaurants reads records from the Restaurants table.
- LIKE '%Cafe' matches names that end with the exact text 'Cafe'. The % wildcard allows any characters before Cafe.
- Example matches: 'Sunrise Cafe' and 'Green Leaf Cafe'. A name like 'Cafe Central' does not match because it does not end with Cafe.

Note: This query assumes the restaurant-name column is called restaurant_name. If your table uses name instead, replace restaurant_name with name.

Task 2 — Products priced from ₹500 to ₹1,500

Question: In a Flipkart-style Products table, use the BETWEEN operator to select all products with a price between 500 and 1500 rupees.

SQL Answer

```
SELECT *
FROM Products
WHERE price BETWEEN 500 AND 1500;
```

Explanation

- SELECT * displays all columns for the products that match the condition.
- BETWEEN 500 AND 1500 selects prices in the range from 500 to 1500.

- The range is inclusive: products priced exactly 500 or exactly 1500 are included.

Note: This query assumes the price column is called price.

Task 3 — Users from selected cities

Question: Write an SQL query to display all users from a Users table whose city is either 'Ahmedabad', 'Surat', or 'Vadodara' using the IN operator.

SQL Answer

```
SELECT *
FROM Users
WHERE city IN ('Ahmedabad', 'Surat', 'Vadodara');
```

Explanation

- IN checks whether city matches any value in the list.
- The query returns users whose city is Ahmedabad, Surat, or Vadodara.
- It is a concise alternative to writing several city comparisons joined with OR.

Task 4 — Artists whose names contain “ar”

Question: Given a table called Songs with columns song_name and artist_name, find all songs where the artist_name contains the letter sequence 'ar' anywhere in the name using the LIKE operator. Hint: Use wildcards on both sides of the pattern.

SQL Answer

```
SELECT song_name, artist_name
FROM Songs
WHERE artist_name LIKE '%ar%';
```

Explanation

- SELECT song_name, artist_name displays the song title and artist name.
- The pattern '%ar%' finds the sequence 'ar' anywhere in artist_name.
- The % before ar allows any characters before the sequence; the % after ar allows any characters after it.
- Example matches include 'Arijit Singh' and 'Charlie Puth', because their artist names contain the sequence 'ar' (case matching depends on the database collation).

Quick Revision

Operator / Pattern	Purpose	Example
LIKE '%Cafe'	Ends with Cafe	name LIKE '%Cafe'
BETWEEN 500 AND 1500	Inclusive price range	price BETWEEN 500 AND 1500
IN (...)	Matches any value in a list	city IN ('Ahmedabad', 'Surat', 'Vadodara')
LIKE '%ar%'	Contains ar anywhere	artist_name LIKE '%ar%'